

The Director LLM: A Multi-Critic Reinforcement Learning Framework for Domain-Aware Narrative Generation

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Abstract

Large Language Models (LLMs) demonstrate remarkable fluency in text generation, yet they struggle with long-term coherence, stateful consistency, and balancing multiple competing objectives. We introduce the **Director LLM**, a novel agent trained via Multi-Critic Reinforcement Learning (MCRL) to act as a domain-aware Dungeon Master in collaborative storytelling. By decomposing DM responsibilities into automatable skills—narrative quality, causal responsiveness, world consistency, and character voice—and training with specialized critics using dynamic reward weighting, the Director achieves improved coherence and controllability. Our framework positions MCRL as a general paradigm for principled, domain-specific generation with applications beyond entertainment to trustworthy AI systems requiring multi-constraint satisfaction.

1 Introduction and Motivation

Large Language Models (LLMs) struggle in domains requiring long-term coherence, stateful consistency, and balancing competing objectives. Creating an AI Dungeon Master (DM) illustrates these challenges: a DM must simultaneously act as storyteller, referee, world simulator, and NPC actor. Standard supervised fine-tuning approaches mimic DM style but fail at deeper responsibilities, suffering from logical contradictions and narrative amnesia.

We introduce the **Director LLM**, trained via Multi-Critic Reinforcement Learning (MCRL) to decompose the DM role into explicit, automatable skills. Our thesis: multi-objective reinforcement learning with specialized critics creates more coherent, consistent interactive narrative agents than traditional supervised approaches.

2 Related Work

Our work synthesizes three key research areas. First, we extend Reinforcement Learning from AI and Human Feedback (Ouyang et al., 2022; Lee et al., 2023), which typically uses monolithic reward models. Recent multi-objective alignment research (Williams et al., 2024; Li et al., 2025) shows that decomposing rewards into explicit aspects leads to better-balanced and more robust models. Second, we build upon grounded language learning in text-based games like TextWorld (Côté et al., 2018) and Jericho (Hausknecht et al., 2020), extending beyond task completion to focus on narrative quality and coherence. Third, our dynamic reward weighting mechanism provides sophisticated controllable generation through context-aware priority adaptation during training.

3 Methodology: Multi-Stage Pipeline

Our methodology is a multi-stage pipeline designed to systematically build and refine the agent’s capabilities. It begins with foundational language and format learning, proceeds to the independent development of skill-specific critics, and culminates in a sophisticated reinforcement learning loop.

3.1 Stage 1: Supervised Component Development

This initial stage creates the supervised models that serve as the starting point for the RL process.

The Core Agent (The "Director" LLM - SFT Baseline):

- **Base Model:** meta-llama/Llama-2-7b-hf. This 7-billion-parameter model is chosen for its strong reasoning capabilities and manageable size.

- **Training Technique:** QLoRA (Dettmers et al., 2023) for parameter-efficient fine-tuning.
- **Training Corpus:** Mixed dataset including the Critical Role D&D Dataset (CRD3) (Ramesh Kumar and Bailey, 2020) and the LIGHT dataset (Urbanek et al., 2019).

The Hybrid Simulated Player:

- **Core Player Model:** distilgpt2, fine-tuned on player utterances from CRD3 and LIGHT datasets.
- **Overlay Classifier Model:** distilbert-base-uncased classifier trained to label player output with discrete intents (EXPLORE, ACTION, DIALOGUE).

The Critics Panel (Reward Models):

- **Narrative Quality Critic:** microsoft/deberta-v3-base model fine-tuned on the ROCStories/Story Cloze corpus (Mostafazadeh et al., 2016).
- **Causal Responsiveness Critic:** Pre-trained NLI model (MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli) used zero-shot to calculate entailment probability.
- **World Consistency Critic:** Programmatic module combining a Python WorldStateTracker class with google/flan-t5-large few-shot intent parser.
- **Character Voice Critic:** microsoft/deberta-v3-base model fine-tuned on CRD3 dataset for character consistency verification.

3.2 Stage 2: The MCRL Fine-Tuning Pipeline

Algorithm: Proximal Policy Optimization (PPO) (Schulman et al., 2017), implemented via Hugging Face TRL with three stability enhancements: KL-divergence penalty against the SFT baseline, reward normalization (z-scoring), and reward clipping.

The Automated Loop: The SFT agent (policy) interacts with the Hybrid Simulated Player for a fixed number of turns per episode. The Director Agent generates a narrative turn, and the Critics Panel scores this turn, producing a reward vector $R_{vec} = [r_{narrative}, r_{causality}, \dots]$. The Hybrid Simulated Player generates a response and clean intent label I . The Dynamic Weighting module

maps the intent label to a pre-defined weight vector, and a final scalar reward R is computed via dot product: $R = R_{vec} \cdot W$. The (prompt, response, reward) tuple is stored in a replay buffer, and PPO samples from the buffer to update the Director Agent’s policy.

4 Evaluation Plan and Metrics

Our evaluation is fully automated, zero-shot, and designed to rigorously test the agent’s acquired skills through comprehensive intrinsic and extrinsic measures.

4.1 Baselines for Comparison

- **SFT Baseline:** QLoRA-tuned LLaMA-7B model without RL.
- **Reranking Agent:** SFT Baseline generating $k=5$ responses, with Narrative Critic choosing the best.
- **RL (Single-Critic):** Agent trained with PPO using single, averaged reward signal.
- **RL (Fixed Weights):** Agent trained with all four critics but static, uniform weights.
- **The Director (Full MCRL):** Our final, complete model with dynamic weighting.

4.2 Intrinsic Evaluation

Average scores from each of the four critics and perplexity on a held-out set from the CRD3 dataset to confirm RL training success and analyze trade-offs made by each model variant.

4.3 Extrinsic Evaluation (Zero-Shot Transfer)

- **Interactive Game Performance:** Pre-defined subset of classic games from Jericho suite (Zork, Adventure). Metric: Average normalized game score.
- **Narrative & Commonsense Coherence:** Story Cloze Test and HellaSwag benchmark (Zellers et al., 2019). Metrics: Accuracy on multiple-choice tasks.
- **Logical Consistency & Grounding:** Programmatic suite of ”challenge probes” to test for specific failure modes like item hallucination and world state amnesia. Metric: Accuracy on targeted tests.

5 Timeline and Implementation

Month 1: Component development and integration. Parallel development of SFT Agent, Simulated Player, and critics, followed by PPO loop integration. Deliverable: Trained Single-Critic PPO Agent.

Month 2: MCRL training and evaluation. Run full MCRL experiments with dynamic weighting, conduct comprehensive evaluation, and analyze results. Deliverable: Final Director model with complete results.

6 Extensions and Future Work

Advanced Critics for Deeper Narrative Intelligence could include a "Dramaturg" Critic for pacing and tension trained on story corpora to reward semantic novelty and adherence to dramatic structures, and a "Lore Keeper" Critic using RAG-based retrieval for canon consistency within established universes. Advanced RL mechanisms could involve Learning the Director's Policy through meta-RL agents that learn optimal reward weights dynamically, and Adaptive Player Modeling where the Simulated Player learns individual human preferences. The framework could expand to Goal-Oriented Generation taking high-level plot goals and Multi-Agent Storytelling enabling cooperative or adversarial AI interactions.

7 Broader Impact

The MCRL framework provides a general methodology for building principled, controllable AI systems in high-stakes domains requiring multi-constraint satisfaction. Applications include Trustworthy AI with critics for factual consistency, logical soundness, and citation quality creating verifiable research assistants; Safe & Ethical AI through scalable alignment with critics for harmlessness, bias mitigation, and privacy adherence; and Domain-Specific Professional Assistants in law (legal precedent, document consistency), medicine (clinical guidelines, HIPAA compliance), and software engineering (correctness, security, readability) with specialized compliance verification.

8 Conclusion

We present the Director LLM, a novel Multi-Critic Reinforcement Learning framework that moves beyond surface-level mimicry to instill underlying

principles of good storytelling. By decomposing the complex role of a Dungeon Master into explicit, automatable skills and training via multi-objective RL, we create a more coherent, consistent, and capable interactive narrative agent with broader applications to principled AI systems.

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